CPU vs GPU vs TPU

27 July 2025 19:57

CPU # Central Processing Unit

- general purpose processor in every computer
- optimized for single threaded or moderately parallel operations.
- Excels in low-latency tasks and handling control logic.

Use-cases: Browsing internet, using office products, small-scale data processing, light ML models.

Feature	Description
Cores	Few (2–32) but very powerful
Best for	General computing, light ML tasks, logic-heavy operations
Latency	Very low
Parallelism	Low
Memory Access	Fast and flexible
Cost	Cheapest

Example of Common Core Configurations:

- **Dual-Core**: 2 cores (e.g., Intel Core i3)
- **Quad-Core**: 4 cores (e.g., Intel Core i5, AMD Ryzen 5)
- **Hexa-Core**: 6 cores (e.g., Intel Core i7, AMD Ryzen 5/7)
- **Octa-Core**: 8 cores (e.g., Intel Core i7/i9, AMD Ryzen 7/9)
- **Deca-Core**: 10 cores (e.g., Intel Core i9, AMD Ryzen 9)
- **High-End Multi-Core**: Up to 64 cores (e.g., AMD Ryzen Threadripper, Intel Xeon)

Limitations

- slower at massive parallelism.
- limited no. of cores for parallel tasks.

GPUs: Graphics Processing Unit

- designed for graphics/video editing needs in early days.
- helps to do parallel processing
- Originally made for rendering graphics, now being

- Originally made for rendering graphics, now being used for AI, ML, DL, LLMs etc.
- GPUs are optimized for throughput over latency.

[Movie recommendation: INTERSTELLAR]

Feature	Description
Cores	Thousands of lightweight cores
Best for	Matrix operations, large ML model training or DL or LLMs
Latency	Moderate
Parallelism	Very high
Memory Access	High bandwidth, but limited flexibility
Cost	Moderate to high

Use-cases: Training neural network models, image processing, video editing & rendering.

→ Apple

- by APC

- Massive parallelism - ideal for matrix multiplications, vector operations
- High memory bandwidth.

TPU : Tensor Processing Units

- specialized hardware built by Google just for TensorFlow framework to practice deep learning at scale
- custom ASIC (Application specific integerated circuit)

- specialized in Tensorflow operations
- extremely fast at matrix-heavy deep learning workloads.

Feature	Description
Cores	Designed specifically for tensor computations
Best for	TensorFlow-based DL workloads (training + inference)
Latency	Very low for tensor ops
Parallelism	Ultra high
Memory Access	Optimized for tensor operations
Cost	Pay-as-you-go (Google Cloud), not for local use

TF

Google cloud Platform (GCP)

→ Forecasting Module.

→ Build LLM models ??

NVIDIA GPUs

TPUs

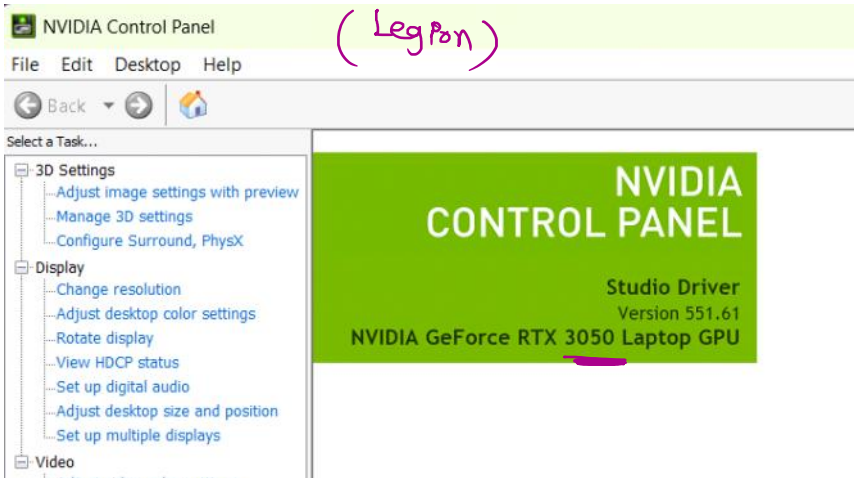
Intermediate machine config.

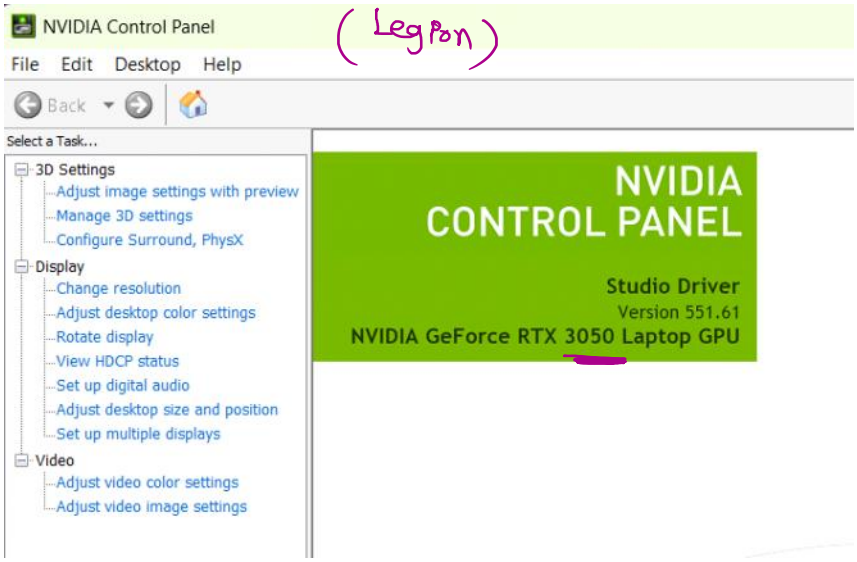
CPU → 6+ cores (i7)

RAM → 32 GB

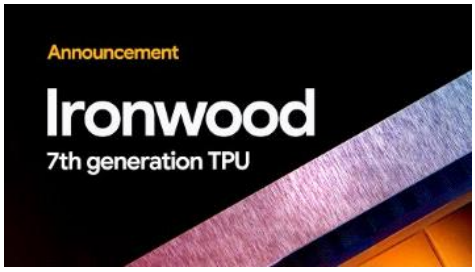
GPU (essential) → NVIDIA RTX 3060 | 4060 (8-12 GB vRAM)

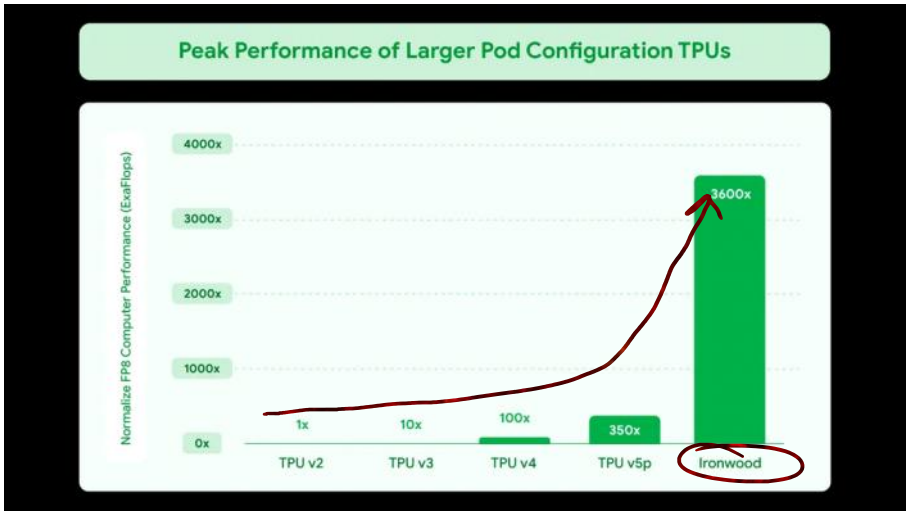
SSD — 1 TB





[Introducing 7th Generation TPUs: Ironwood]





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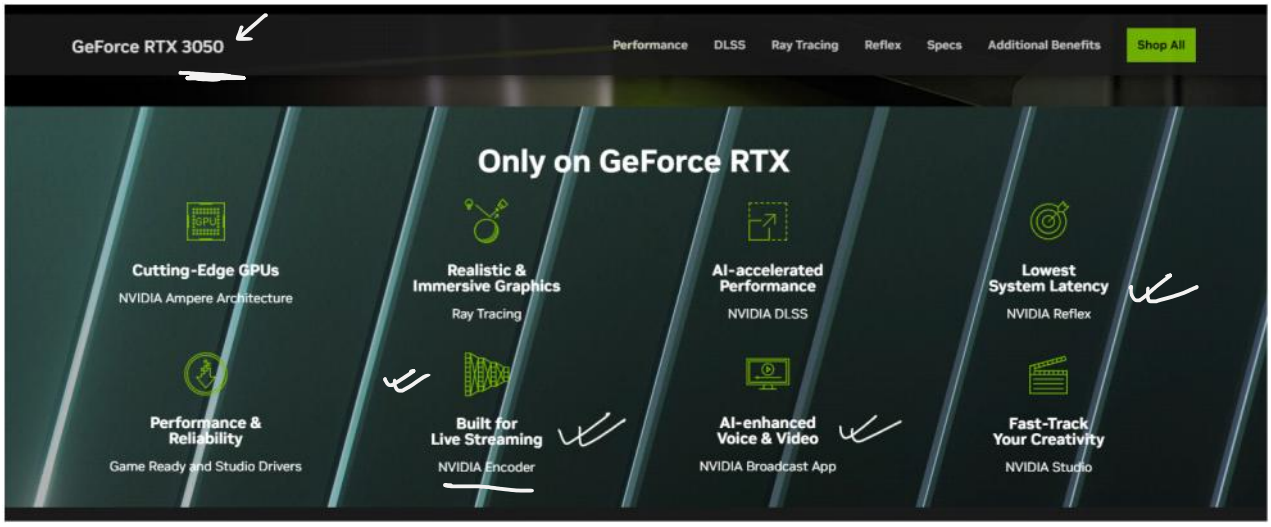
8GB vRAM → 40K



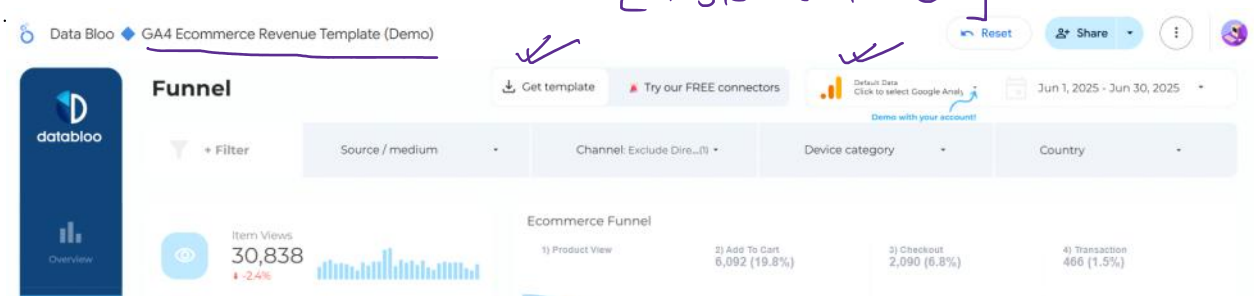
Comparing CPU vs GPU vs TPU

Feature	CPU	GPU	TPU
Purpose	General-purpose	Parallel computing	Deep learning acceleration
Cores	Few (2–64)	Thousands (small cores)	Matrix Units (MXUs)
Parallelism	Low–medium	High	Ultra-high
Flexibility (Memory Access)	Very high	Moderate	Low
Precision	High (int64/float64)	Medium (float32/float16)	Lower (bfloat16/int8)
Use case fit	OS, apps, logic	DL, rendering, ML	TensorFlow models (DNNs)
Programming languages	C/C++, Java, Python, etc.	CUDA, OpenCL, PyTorch, etc.	TensorFlow, JAX
Power efficiency	Medium	Low–medium	High
Accessibility	Ubiquitous	Widely available (NVIDIA, AMD)	Google Cloud only

Looker Studio

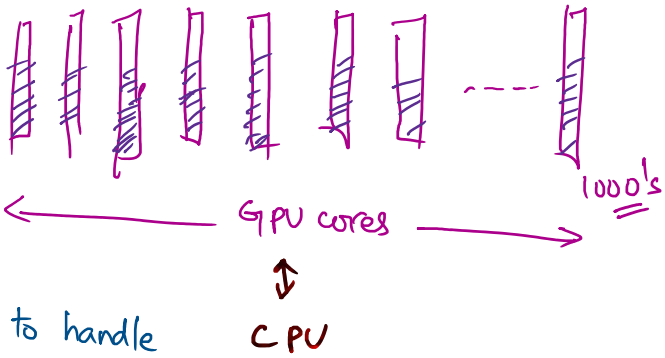
https://lookerstudio.google.com/u/0/reporting/0dcc27d8-13a8-4025-b9dd-a09e1dd78bf1/page/p_kbsk8c7m0c

[Google Looker Studio]



What does "Flexibility for Memory Access" mean ??

- It refers to how easily and efficiently a system (CPU/ GPU) can:
  - access different parts of memory



- Idea is to handle large datasets without crashing.

Latency and Throughput

office of P.O.

No. of letters → Letter Post-Office is processing / day is Throughput

Hotstar

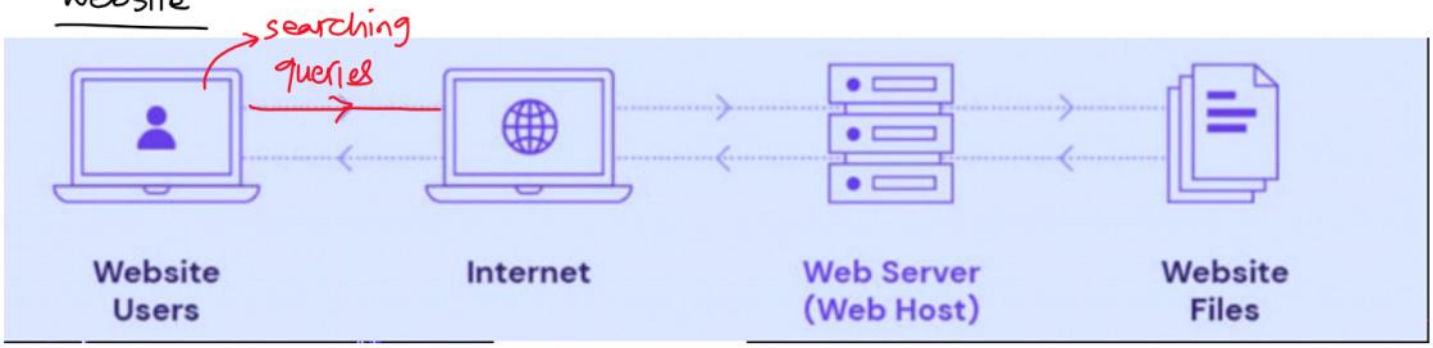
high

Latency is the delay in transmission time

4.2 cr

throughput is like how many viewers are watching the match concurrently

Website



on an avg. → how much time is taken to respond a query → LATENCY

→ how many queries are being resolved/day → THROUGHPUT

IRCTC website

→ getting a ticket is getting an award.

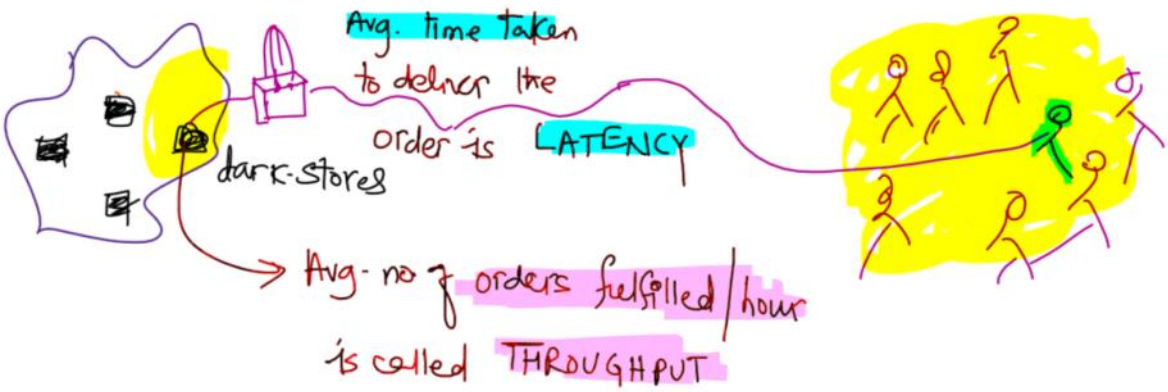
Tatkal period → login to the system → latency is super high.

→ thousands of potential passengers trying to book a ticket in the same train

↓

server load is high.

Zepto | Blinkit (Quick commerce)



LOW LATENCY & HIGH THROUGHPUT

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